

Java Assignment 3

Question 1: Design a College ERP (Student Module)

Requirements:

- Each student has:
 - Roll number, name
 - Marks in multiple subjects
 - Contact details (phone, email)
 - System should:
 - Store multiple students
 - Calculate total & average marks
 - Track total students admitted
 - Support:
 - Flexible average calculation
 - Display formatted student report
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Question 2: Design a Fleet Management System

Requirements:

- Base class: Vehicle
 - Derived classes: Car, Bike
 - System should:
 - Track brand & speed
 - Store fuel type / mileage
 - Use:
 - Static block → system initialization
 - Non-static block → object creation log
 - Allow:
 - Updating vehicle speed
 - Display vehicle details
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Question 3: Design a Banking System

Requirements:

Each account has:

- Account number, holder name
- Balance

System should:

- Store multiple accounts
- Deposit & withdraw money
- Prevent withdrawal if balance is insufficient
- Track total number of accounts

Support:

- Overloaded methods for deposit (cash / online)

Display:

- Account details with updated balance
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Question 4: Design a Hospital Management System

Requirements:

Each patient has:

- Patient ID, name, age
- Disease
- Contact details

System should:

- Store multiple patients
- Identify senior citizens (age ≥ 60)
- Count total patients

Support:

- Use object composition (Patient $\rightarrow$ ContactDetails)

Display:

- Patient details with status
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Question 5: Design a Product Inventory System

Requirements:

Each product has:

- Product ID, name
- Price, quantity

System should:

- Store multiple products
- Calculate total inventory value
- Update stock quantity

Support:

- Use constructor for initialization
- Use method overloading for price update (discount / direct update)

Display:

- Product details and stock value
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Question 6: Design a Library Management System

Requirements:

Each book has:

- Book ID, title, author
- Availability status

System should:

- Store multiple books
- Issue and return books

- Prevent issuing unavailable books

Support:

- Track total books

Display:

- Available and issued books
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Question 7: Design an Employee Management System

Requirements:

Each employee has:

- Employee ID, name
- Salary

System should:

- Store multiple employees
- Calculate salary with allowances
- Track total employees

Support:

- Use method overloading for salary calculation

Display:

- Employee salary details
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Question 8: Design an Online Order System

Requirements:

Each order has:

- Order ID, customer name
- Items and price

System should:

- Store multiple orders
- Calculate total bill
- Apply GST

Support:

- Use array for multiple items
- Use constructor for initialization

Display:

- Final bill with details
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Question 9: Design a Course Management System

Requirements:

Each course has:

- Course ID, course name
- Fees

Each student:

- Can enroll in courses

System should:

- Store students and courses
- Track enrollments
- Count total courses

Display:

- Student course details
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Question 10: Design a Transport Ticket System

Requirements:

Each ticket has:

- Ticket ID, passenger name
- Source, destination
- Fare

System should:

- Store multiple bookings
- Calculate total fare
- Update fare

Display:

- Ticket details